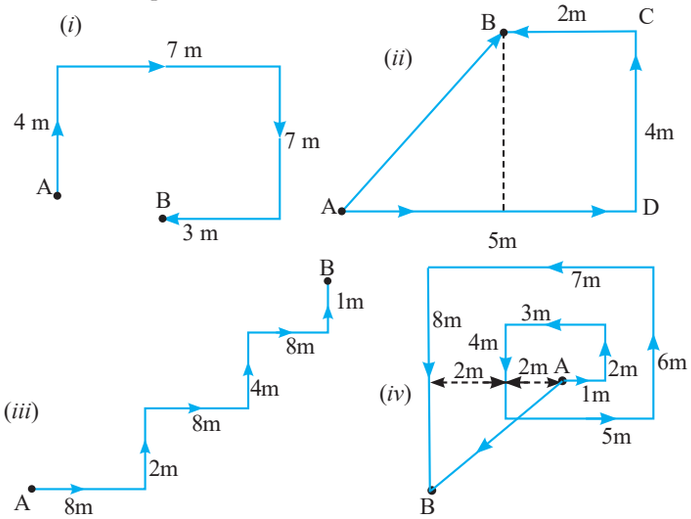
4 Motion in 1-D

Distance and Displacement

1. Find displacement between A & B



2. Correct statement among the following is.

(*a*) When displacement is zero, distance traveled is not zero.

(*b*) When displacement is zero, distance traveled is also zero.

(*c*) When distance is zero, displacement is not zero.

(*d*) Distance traveled and displacement are always equal

(*e*) None of the above statement is correct

3. Fill in the blanks : (may/must/must not/may not)

(*a*) Distance traveled by object is zero, the displacement .....be non-zero.

(*b*) Distance traveled by object is not zero then displacement ..... be zero.

(*c*) Displacement moved is zero then distance .....or ..... zero

(*d*) If displacement is not equal to zero, then distance ..... equal to zero

4. Particle is moving on straight line, distance is equal to displacement

(*a*) True (*b*) False

5. Displacement independent of choice of frame of reference

(*a*) True (*b*) False

6. For a particle moving on parabolic path distance always grater than displacement

(*a*) True (*b*) False

7. Displacement is not zero; then distance must not be zero

(*a*) True (*b*) False

8. Distance is not equal to zero; then displacement may be equal to zero

(*a*) True (*b*) False

9. **Statement-I:** Distance and displacement is different physical quantity.
Statement-II: Distance and displacement have same unit and dimension.

(*a*) Both statement false

(*b*) Both statement true

(*c*) Statement-I is false and Statement-II is true

(*d*) Statement-I is true and Statement-II is false

10. **Statement-I :** Rest & Motion are absolute terms.
Statement-II : Distance can't be decrease with time while displacement can be.

(*a*) Both statement false

(*b*) Both statement true

(*c*) Statement-I is false and Statement-II is true

(*d*) Statement-I is true and Statement-II is false

11. Which of the following option is correct for motion in 1-D, without change in direction.

(*a*) Distance = |displacement |

(*b*) Distance > |displacement |

(*c*) Distance ≥ |displacement |

(*d*) Distance < |displacement |

12. Which of the following option is correct for motion in 1-D, with change in direction.

(*a*) Distance = |displacement |

(*b*) Distance > |displacement |

(*c*) Distance ≥ |displacement |

(*d*) Distance < |displacement |

13. Which of the following option is never correct for a moving object?

(*a*) Distance = |displacement |

(*b*) Distance > |displacement |

(*c*) Distance ≥ |displacement |

(*d*) Distance < |displacement |

14.

(*i*)	Object is moving in straight line	(A)	Distance ≤ displacement
(*ii*)	Object is moving on straight line Without change in direction	(B)	Distance > displacement
(*iii*)	Object is Projected upward from ground	(C)	Distance = displacement
(*iv*)	Moving on circular Path	(D)	Distance ≥ displacement

15. The numerical ratio of displacement to distance is..

(*a*) always less than 1 (*b*) always greater than 1

(*c*) always equal to 1 (*d*) may be less than 1 or equal to 1

16. If x denotes the ratio of displacement to distance, the possible value of x is :

(*a*) 3 (*b*) 6/7

(*c*) 1 (*d*) Both (*b*) and (*c*)

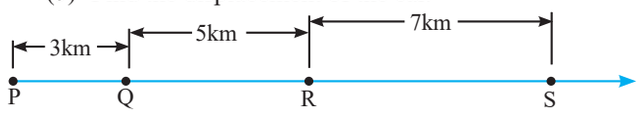
17. If initial position of object (2, 6, 9) and final position (8,–2, 19) then find displacement and distance

18. Object moves from A(2, 3, –4) to B(3,4,1) to C(5,7,1). Find distance and displacement from A to C.

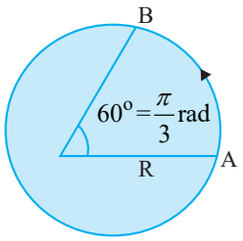
19. A car moving along in a straight highway from point P to point Q to point R and to point S, then back to point Q and finally to the point R as shown in the figure below.

(*a*) Find the distance travelled by car.

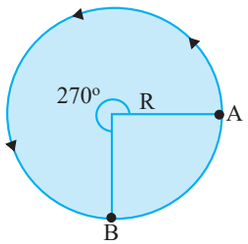
(*b*) Find the displacement of the car.



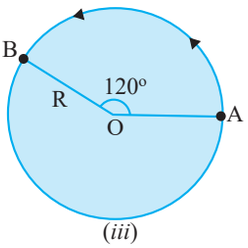
20. Object moves on a circular path by angle 60° then find ratio of distance to displacement.



(*i*)

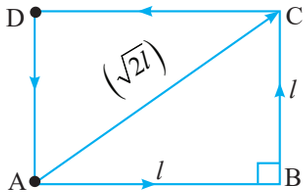


(*ii*)



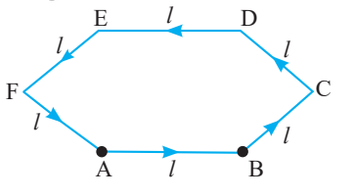
(*iii*)

21. An object is moving along the sides of a square park with a side length of ℓ. The motions of the object between points A, B, C, and D are listed in the given figure. For each motion, match the following:



Motion		Displacement		Distance	
1.	A→B	a.	ℓ	i.	3ℓ
2.	A→C	b.	ℓ	ii.	4ℓ
3.	A→D	c.	0	iii.	ℓ
4.	A→A	d.	$\sqrt{2\ell}$	iv.	2ℓ

22. An object is moving along the sides of a regular hexagonal path with side length ℓ. For each motion:



Match the corresponding displacement travelled by the object

Motion		Displacement	
1.	A → B	a.	2ℓ
2.	A → C	b.	0
3.	A → D	c.	$\sqrt{3}\ell$
4.	A → E	d.	ℓ
5.	A → F	e.	3ℓ
6.	A → A		

23. Match the corresponding distance travelled by the object:

Motion		Displacement	
1.	A → B	a.	5ℓ
2.	A → C	b.	2ℓ
3.	A → D	c.	6ℓ
4.	A → E	d.	4ℓ
5.	A → F	e.	3ℓ
6.	A → A	f.	ℓ

24. Kallua is moving on a circular path and completes one full round in 40 seconds. After moving for 3 minutes and 20 seconds, what will be Kallua's displacement?

25. The position of an object is given by the equation: $x = (t^2 + 2t + 3)m$ Find the position of the object at $t = 2$ seconds and displacement in 2 sec.

26. A student moves as follows: 10 meters towards the east, then 20 meters towards the south, and then $20\sqrt{2}$ meters in the northwest direction. Find the displacement of the student from the starting point.

27. Ramlal Moves 10 m North then 10 m east, then 10 m climb upward then find his displacement.

28. A person moves 20 m North then 30 m towards East and finally $40\sqrt{2}$ m South-West then his displacement.

(*a*) $10\sqrt{2}\tan^{-1}(2)$ South of west

(*b*) $10\sqrt{5}\tan^{-1}(2)$ South of west

(*c*) $10\sqrt{5}\tan^{-1}(1/2)$ West of South

(*d*) Both (*a*) and (*b*)

29. A butterfly start flying from a corner of the cubical room of side *l* and reaches to the opposite corner of the room. Find its displacement.

30. An ant start moving from a corner of the solid cubical room and want to reach opposite of body diagonal) find (minimum distance moved by ant.

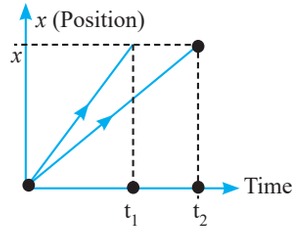
31. A carom board (4 ft × 4 ft) square) has the queen at the centre. The queen, hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line. Find the magnitude of displacement of the queen

(*i*) from the centre to the front edge

(*ii*) from the front edge to the hole and

(*iii*) from the centre of the hole.

32. Position-time graph for two man are moving along *x*-axis as shown; then find correct relation between



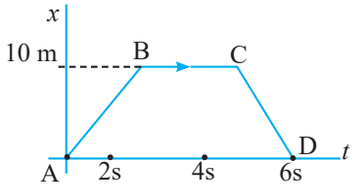
(*a*) Displacement = distance (*b*) Displacement > distance

(*c*) Displacement < distance (*d*) None of these

33. For given between position and time graph.

(*i*) Calculate the total distance traveled by the object from point A to D.

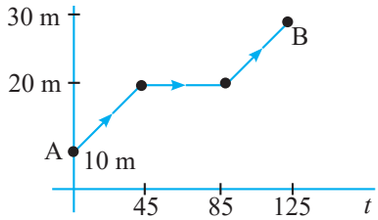
(*ii*) Calculate the displacement of the object from point A to D.



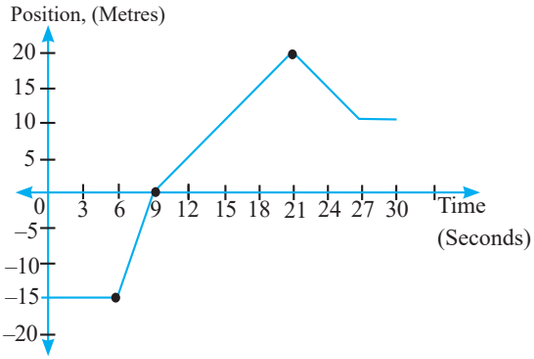
34. An object moves along a path according to the position-time graph is shown.

(*i*) Calculate the distance traveled by the object from A to B.

(*ii*) Determine the displacement of the object from A to B.



35. The position-time graph for an elevator travels up and down is given below. Find the distance and displacement of the elevator between 6 seconds and 21 seconds.



36. A drunkard is walking along a straight road. He takes 5 steps forward and 3 steps backward and so on. Each step is 1 m long and takes 1 s. There is a pit on the road 11 m away from the starting point. The drunkard will fall into the pit after:

(*a*) 21 s (*b*) 29 s (*c*) 31 s (*d*) 37 s

37. Magnitude of initial position vector is 3 m and its makes 15° angle from x-axis and magnitude of final position vector is 4 m and its makes 75° angle from *x*-axis. Find out displacement of particle.

38. A particle can move only along x-axis. Three pairs of initial and final positions of particle at two successive times are given

Pair	Initial Position	Final Position
1	–3m	+5m
2	–3m	–7m
3	+7m	–3m

Find the sum of magnitudes of displacement in the pairs which give negative displacement in *m*.

(*a*) 14 (*b*) 12 (*c*) 20 (*d*) 22

Speed and Velocity

39. Object is moving in straight line without change in direction, moves x_1 distance in time t_1 and then x_2 distance in time t_2 then find average speed.

40. Object is at position x_1 at time t_1 and at position x_2 at time t_2 then find average speed & average velocity if is it moving without a change in direction.

41. Instantaneous velocity of a particle-

(A) depends on instantaneous position

(B) depends on instantaneous speed

(C) independent of instantaneous position

(D) independent of instantaneous speed

(*a*) Both (A) & (B) are correct

(*b*) Both (C) & (D) are correct

(*c*) Both (A) & (D) are correct

(*d*) Both (B) & (C) are correct

42. Consider the motion of the tip of the minute hand of a clock for one hour motion which of the following is/are correct.

(A) The displacement is zero

(B) The distance covered is zero

(C) The average speed is zero

(D) The average velocity is zero/

Option

(*a*) (A) & (D) (*b*) (B) & (C)

(*c*) (A) & (C) (*d*) (B) & (D)

43. An object moves with speed v_1, v_2 and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where AB = BC and AD = 3AB, then average speed of the object will be:

(*a*) $\dfrac{v_1v_2v_3}{3(v_1v_2+v_2v_3+v_3v_1)}$ (*b*) $\dfrac{v_1+v_2+v_3}{3}$

(*c*) $\dfrac{v_1+v_2+v_3}{3v_1v_2v_3}$ (*d*) $\dfrac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_3v_1}$

44. A Train has speed 60 km/hr for one -hr and 40km/hr for next half-hr, then average speed in km/hr.

(*a*) 50 (*b*) 53.33 (*c*) 48 (*d*) 70

45. Object Moves for 10 s with speed 20 m/s and then it Moves with speed 30 m/s for 10 sec then find average speed for Complete Journey.

46. Object moves 10 m with speed 20 m/s and then it Moves with speed 30 m/s for 10 sec. then find Avg. speed for Complete Journey.

47. Object moves 20 m with speed 20 m/s and then it moves with speed 30 m/s for 20 m then find Avg. speed for Complete Journey.

48. If object moves $\frac{2}{5}$th distance of journey with speed 10 m/s and remaining with 30 m/s then average speed will be :

49. A car travels from Kota to Jaipur with speed 30 km/h, and it returns along the same path with speed 60 km/h. Calculate average speed of the car and Avg. velocity ?

50. A body covers first one-third of the distance with velocity $10ms^{-1}$ in same direction, the second one – third with a velocity $20ms^{-1}$ and last one-third with a velocity of 30 ms^{-1} . The average velocity of body is

(*a*) 17.8 ms^{-1} (*b*) 16.4 ms^{-1}

(*c*) 18.3 ms^{-1} (*d*) 20.2 ms^{-1}

51. Object Moves with V_1 for $t/3$ and with V_2 for $\frac{2t}{3}$ then find Avg. speed

52. A truck moves a distance of 50 km. It covers first half to the distance at speed of 200 m /s and second half at speed v. If average speed of truck is100 m/s then value of v is

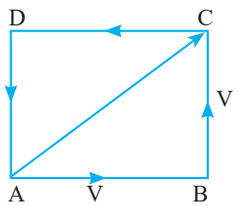
53. A bus travels its half distance of journey with speed 5 m/s. It covers remaining distance in two equal time intervals with speed 15m/s Calculate average speed of the bus for the whole journey.

54. A person travels along a straight road for the first t/3 time with a speed V_1 and for next $2t$ / 3 time with a speed v_2. Then the mean speed v is given by

(*a*) $v=\dfrac{v_1+2v_2}{3}$ (*b*) $\dfrac{1}{v}=\dfrac{1}{3v_1}+\dfrac{2}{3v_2}$

(*c*) $3=\dfrac{1}{3}\sqrt{2v_1v_2}$ (*d*) $v=\sqrt{\dfrac{3v_2}{2v_1}}$

55. Object is moving with constant speed (v) on square track then Fill in the blank for average speed and average velocity.



Motion	Avg. Speed	Avg. Velocity
A → B		
A → C		
A → D		
A → A		

56. If Instantaneous speed is constant, then what about average speed

57. A particle is moving such that its position coordinates (x,y) are (2m, 3m) at time t = 0, (6m,7m) at time t = 2s and (13m, 14 m) at time t = 5s. The average velocity vector $\vec{v}_{avg}$ from t = 0 to t = 5 s

(*a*) $\frac{1}{5}(13\hat{i}+14\hat{j})$ (*b*) $\frac{7}{3}(\hat{i}+\hat{j})$

(*c*) $32(\hat{i}+\hat{j})$ (*d*) $\frac{11}{5}(\hat{i}+\hat{j})$

58. Position $x = t^2-2t+4$ then find, velocity at t = 2sec

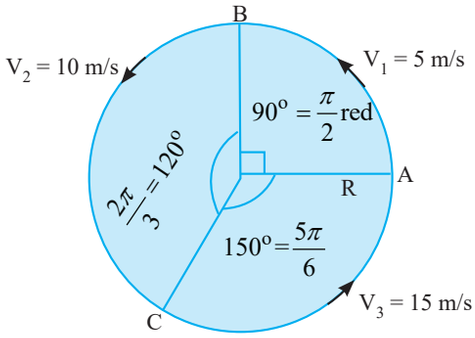
59.

Velocity	A.....	B.....	If velocity is constant then what about speed.	It velocity is variable then what about speed
Speed	It speed is constant then what about velocity ??	If speed is variable then what about velocity	C.....	D.....

60. Position of object $x = 10\ t - 2t^2$. Find time when object comes to at rest.

61. If object is moving with speed $v = 3t^2$; then find average speed in 2-sec.

62. A man is moving on circular path as shown then find avg. speed in one rotation.

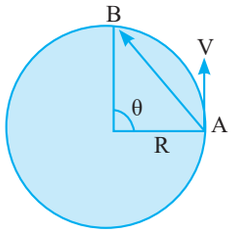


63. Object is moving on circular path with constant speed then find Average Velocity when it completed half revolution.

64. A car runs at a constant speed on a circular track of radius 100m, taking 62.8 seconds for every circular lap. The average speed for half circular lop is

(*a*) 10 m/s (*b*) zero (*c*) 20 m/s (*d*) 30 m/s

65. Object is moving on circular path with speed v then find avg. velocity when it moves an angle θ.



66. A particle is executing a circular motion of radius R with a uniform speed v. After completing half the circle, the change in velocity, and in speed will be respectively

(*a*) zero, zero (*b*) $2v$, zero (*c*) $2v$, $2v$ (*d*) zero, $2v$

67. **Assertion (A):** Object is moving on circular path with constant speed then the change in velocity will be zero when it will complete it's half revolution as it is moving with same speed.
Reason (R): Velocity is vector quantity.

(*a*) Assertion (A) is false but Reason (R) is true

(*b*) B Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

68. **Assertion (A):** The average and instantaneous velocities have same value in a uniform motion.

Reason (R): In uniform motion, the velocity of an object increases uniformly.

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

(*d*) If both Assertion (A) & Reason (R) are false.

69. Average velocity in a time interval zero then in same time interval average speed is:

(*a*) Must be zero (*b*) May be zero

(*c*) Must be non-zero (*d*) may be-ve

70. **Assertion (A):** The average velocity of the object over an interval of time is either Smaller than or equal to the average speed of the object over the same interval.

Reason (R): Velocity is a vector quantity and speed is a scalar quantity.

(*a*) If both Assertion (A) & Reason (R) are True & the Reason (R) is a correct explanation of the Assertion (A).

(*b*) If both Assertion (A) & Reason (R) are True but Reason (R) is not a correct explanation of the Assertion (A).

(*c*) If Assertion (A) is True but the Reason (R) is False.

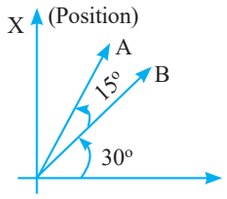
(*d*) If both Assertion (A) & Reason (R) are false.

71. Object is moving with constant speed then velocity of object:

(*a*) may be variable (*b*) must be constant

(*c*) must be variable (*d*) may be zero

72. Find ratio of velocity of A to B



Acceleration

73. Which of the following is possible:

(*a*) $\vec{v} = \cos t^n$ $\vec{a} = \cos t^n$

(*b*) $V\uparrow$ $a = \cos t^n$

(*c*) $V\uparrow$ $a = 0$

(*d*) $V\uparrow$ $a\downarrow$

(e) $V\downarrow$ $a\uparrow$

(f) $V = \cos t^n$ $a\uparrow$

(g) $V = \cos t^n$ $a = \cos t^n$

74. Which of the following option is correct:

(a) Velocity of object increasing and acceleration may decreasing

(b) Velocity of object decreasing and acceleration may increasing

(c) Acceleration may be non-zero when velocity of object is zero.

(d) All of these.

75. **Statement-I :** If acceleration of particle is increasing, it's velocity must increases.

Statement-II : The rate of change of speed with respect to time will provide us Net acceleration.

(a) Both statements false

(b) Both statements true

(c) Statement-I is false and Statement-II is true

(d) Statement-I is true and Statement-II is false

76. Object is moving such that its velocity and acceleration is in opposite direction then

(a) Speed may constant.

(b) Speed may increasing.

(c) Speed must be decreasing.

(d) Speed may be increasing or decreasing:

77. An object is moving with constant velocity then which of the following option is correct

(a) Acceleration may be increasing.

(b) Acceleration is zero.

(c) Acceleration is decreasing.

(d) (Acceleration is non-zero.

78. Which of the following is wrong

(a) Velocity increasing and acceleration decreasing.

(b) $\vec{v}\downarrow$ and $\vec{a}\uparrow$

(c) $\vec{v}\uparrow$ and $\vec{a} = 0$

(d) None of these.

79. Object is moving with acceleration 2 m/s² its velocity at $t = 0$ is 10m/s then find its velocity at $t = 4$s

80. Velocity at $t = 0$ sec is 10 m/s its velocity becomes 40 m/s after 6 sec then find acceleration.

81. Velocity at $t = 2$sec is 20 m/s. At $t = 5$sec, it becomes 32 m/s. Then velocity at $t = 7$sec will be:

82. An object may have

(A) varying speed without having varying velocity

(B) varying velocity without having varying speed

(C) non-zero acceleration without having varying velocity

(D) non-zero acceleration without having varying speed

(a) Only B is correct (b) Only D is correct

(c) Both B and D are correct (d) All are correct

83. The velocity of a particle is zero at $t = 0$ then

(A) the acceleration at t = 0 must be zero

(B) the acceleration at t = 0 may be zero

(C) if the acceleration is zero from $t = 0$ to t = 10s, speed is also zero in this interval

(D) if the speed is zero from t = 0 t = 10 sec, the acceleration is also zero in the interval.

(a) A), (C) and (D) are correct

(b) (B), (C) and (D) are correct

(c) (A) and (D) are correct

(d) (B) and (C) are correct

84. For a body moving on a straight line if x is position coordinate and t is time then acceleration of body is constant when -

(A) x and velocity is linear

(B) x and(square of velocity is linear

(C) t and velocity is linear

(D) t and square of velocity is linear

(a) Both (A) & (B) are correct

(b) Both (C) & (D) are correct

(c) Both (A) & (D) are correct

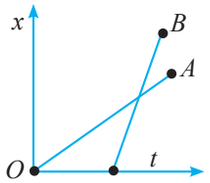
(d) Both (B) & (C) are correct

85. In a car race on straight road, car A takes a time 't less than car B at the finish and passes finishing point with a speed 'V' more than that of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then 'v' is equal to

(a) $\frac{2a_1a_1}{a_1 + a_2}t$ (b) $\sqrt{2a_1a_2}\,t$

(c) $\sqrt{a_1a_2}\,t$ (d) $\frac{a_1 + a_2}{2}t$

86. The position-time graphs for two students A and B returning from the school to their homes are shown in figure:



(A) A lives closer to the school

(B) B lives closer to the school

(C) A takes lesser time to reach home

(D) A travels faster than B

(E) B travels faster than A

Choose the correct answer from the options given below:

(a) (A) and (E) only (b) (B) and (E) only

(c) (A), (C) and (E) only (d) (A), (C) and (D) only

87. Ramlal is moving with velocity $3\hat{i} + 4\hat{j}$ at t = 0 after 5 sec its velocity becomes $4\hat{i} + 3\hat{j}$ then find average acceleration.

88. Kallu is moving with speed 40 m/s in north after 10 sec he is moving with 40 m/s in east then find.

(i) Magnitude of rate of change in velocity.)

(ii) Rate of change in (magnitude of velocity.)

89. Find acceleration in each case.

(*i*) $x = 4t + 6$, (*ii*) $x = 3t^2 + 4t + 6$, (*iii*) $x = 2t^3 + 5t$, (*iv*) $x = t^4 + 4t$, (*v*) $V = 3t^2 + 4$, (*vi*) $V = 3t + 4$, (*vii*) $V = t^3 + 4$, (*viii*) $V = 2x + 4$

90. If $x = \frac{4}{t^2}$ then find acceleration

91. In which case acceleration is non zero constant.

(*i*) $x = t^2 + 2t$ (*ii*) $x = 1/t^2$

(*iii*) $x = et$ (*iv*) $x = t^3$

(*v*) $x = \sin(t)$ (*vi*) $xt^2 = \cos t^n$

(*vii*) $x = \sqrt{3t+5}$ (*viii*) $V = t^2$

(*ix*) $V = kt$ (*x*) $V = \sin(t)$

(*xi*) $V = kx^2$ (*xii*) $V = k\sqrt{x}$

(*xiii*) $V = \frac{1}{x}$

92. Object is moving such that its position given as a function of time $x = \alpha t^2 + \beta t + \gamma$ then find initial velocity, initial acceleration and initial position.

93. Object is moving with velocity $V = 4t^2 + 2t + 4$ then find velocity and acceleration at $t = 1$ sec.

94. If $V = kx$ the find acceleration at $x = 2$m.

95. Velocity of object $V = 3t^2$ then find acceleration in 2 sec and acceleration at 2 sec.

96. If position $x = t^2 + 5t^3 + 6$ then find

(*i*) Initial acceleration.

(*ii*) Initial velocity.

(*iii*) Acceleration at $t = 2$sec.

97. If position $x = at^2 - bt^3$ find time when the acceleration is zero.

98. If position of object $x = t^2 - 4t + 5$ then find instant when velocity becomes zero and displacement when object comes to at rest.

99. If velocity of object $V = 3t^2$ then find distance in 2 sec .

100. The position of a particle moving along X -axis in given by $x = 10t - 2t^2$. Then the time (t) at which it will momentarly come to rest is

(*a*) 0 (*b*) 2.5 s (*c*) 55 (*d*) 10 s

101. If the displacement of a particle varies with time as $\sqrt{x} = t + 7$, then

(*a*) Velocity of the particle is inversely proportional t

(*b*) Velocity of the particle is proportional to t^2

(*c*) Velocity of the particle is proportional to $\sqrt{t}$

(*d*) The particle moves with constant acceleration

102. The position x of particle moving along x-axis varies with time t as x = A sin (ωt) where A and w are positive constants. The acceleration a of particle varies with its position (x) as

(*a*) $a = Ax$ (*b*) $a = -\omega^2 x$

(*c*) $a = A\,\omega\,x$ (*d*) $a = \omega^2 xA$

103. The initial velocity of a particle is u (*at* $t = 0$) and the acceleration a is given by $at^{3/2}$. Which of the following relations is valid?

(*a*) $v = u + \alpha t^{3/2}$ (*b*) $v = u + \frac{3\alpha t^3}{2}$

(*c*) $v = u + \frac{2}{5}\alpha t^{5/2}$ (*d*) $v = u + \alpha t^{5/2}$

104. A particle of unit mass undergoes one-dimensional motion such that its velocity varies according to $v(x) = bx^{-2n}$, where b and n are constants and x is the position of the particle. The acceleration of the particle as a function of x, is given by

(*a*) $-2n\beta^2 e^{-4n+1}$ (*b*) $-2n\beta^2 x^{-2n-1}$

(*c*) $-2n\beta^2 x^{-4n-1}$ (*d*) $-2\beta^2 x^{-2n+1}$

105. If velocity $V = k\sqrt{x}$ then find acceleration.

106. If velocity $v \propto \sqrt{x}$ then which of the following function is correct for position time relation.

(*a*) $x \propto t$ (*b*) $x \propto t^2$

(*c*) $x \propto \sqrt{t}$ (*d*) $x \propto t^{3/2}$

107. The velocity of a body depends on time according to the equation $v = (t + 6)^2$ The body is undergoing.

(*a*) Uniform motion

(*b*) Non-uniform motion with uniform acceleration.

(*c*) Non-uniform motion with non-uniform acceleration.

(*d*) None of the above.

108. The position of a particle moving along the x-axis at certain times is given below:

t (*s*)	0	1	2	3
x (*m*)	-2	0	6	16

Which of the following describes the motion correctly?

(*a*) Uniform: accelerated

(*b*) Uniform, decelerated

(*c*) Non-uniform, accelerated

(*d*) There is not enough data for generalization.

109. A particle moves along a straight line such that its displacement at any time t is given by $s = (t^3 - 6t^2 - 3t + 4)$ meters. The velocity when the acceleration is zero is

(*a*) 3 m/s (*b*) 42 m/s (*c*) -9 m/s (*d*) -15 m/s

110. Position of object $x = t^3 - 6t^2 + 10$ find time when acceleration of object will be zero.

111. Position of object $x = 2t^3 - 4t^2 + 4$ then find, velocity acceleration and position at $t = 2$sec

112. Position $x = t^3 - 9t^2 + 4$ Find velocity when acceleration is zero.

113. Position of object $x = 2t^3 - 24t + 4$ find acceleration, when velocity of object is zero.

114. A particle moving along x-axis has acceleration f, at time t, given $f = f_0\left(1 - \frac{t}{T}\right)$, where f_0 and T are constants. The particle at $t = 0$ has zero velocity. At the instant (t = T) the particle's velocity is

(*a*) $\frac{1}{2}f_0 T$ (*b*) f_0T (*c*) $\frac{1}{2}f_0 T^2$ (*d*) f_0T^2

115. A body is moving with variable acceleration (a) along a straight line. The average acceleration of body in time interval t_1 to t_2 is

(*a*) $\frac{a[t_2+t_1]}{2}$ (*b*) $\frac{a[t_2-t_1]}{2}$

(*c*) $\frac{\int_{t_1}^{t_2} adt}{t_2+t_1}$ (*d*) $\frac{\int_{t_1}^{t_2} adt}{(t_2-t_1)}$

116. A particle moves in a straight line and its position x at time t is given by $x^2 = 2 + t$. Its acceleration is given by

(*a*) $\frac{-2}{x^3}$ (*b*) $-\frac{1}{4x^3}$ (*c*) $-\frac{1}{4x^2}$ (*d*) $\frac{1}{x^2}$

117. A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$ The acceleration of particle is proportional to

(*a*) (Velocity) $^{3/2}$ (*b*) (Distance)^2

(*c*) (Distance)$^{-2}$ (*d*) (Velocity)$^{2/3}$

118. If acceleration of object $a = 2x^{3/2}$ then find velocity at x where initial $x = 0$ is 4 m/s

119. In which option object is moving in one-dimension without change in direction. ??

(*a*) $x = t^2 + 6t + 4$ (*b*) $x = -t^2 - 4t + 4$

(*c*) $x = 4t - t^2 + 5$ (*d*) Both (*a*) and (*b*)

120. Position $x = t^2 + 6t + 4$ then find distance and displacement in $t = 4$ sec.

121. If position $x = t^2 - 6t + 2$ then find distance and displacement in 4 sec.

122. If position of object $t^2 - 4t + 3$ then find avg. speed (total distance) and avg. velocity (total distance) in 4 sec.

123. If one – dimensional motion, instantaneous speed v satisfies $0 \le v < v_0$.

(*a*) The displacement in time T must always take non - negative values

(*b*) The displacement x in time T satisfies with variable $-v_0T < x < v_0T$

(*c*) The acceleration is always a non-negative number

(*d*) The motion has no turning points

Motion in Straight Line with Constant Acceleration

124. If a car at rest accelerates uniformly to a speed of 144 km/h in 20 s, it covers a distance of

(*a*) 1440 cm (*b*) 2980 cm (*c*) 20 m (*d*) 400 m

125. The velocity of the bullet becomes one third after it percentages 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling of $(4 + x)$ cm inside the block. The value of x is:

(*a*) 2 (*b*) 1 (*c*) 0.5 (*d*) 1.5

126. Object starts from rest and constant acceleration moves 80 m in 7 sec. then find displacement in next 7 sec.

127. A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 m/s to 20 m/s while passing through a distance 135 m in t second. The value of t is

(*a*) 9 (*b*) 10/9 (*c*) 10 (*d*) 90

128. The velocity of a body moving with a uniform acceleration of 2 m/sec² is 10 m/sec. Its velocity after an interval of 4 sec is

(*a*) 12 m /sec (*b*) 14 m /sec

(*c*) 16 m /sec (*d*) 18 m /sec

129. A body of mass 10 kg is moving with a constant velocity of 10 m/s. When a constant force acts for 4 seconds on it, it moves with a velocity 2 m/sec in the opposite direction. The acceleration produced in it is:

(*a*) 3 m /sec^2 (*b*) –3 m /sec^2

(*c*) 0.3 m /sec^2 (*d*) –3 m /sec^2

130. Object starts his motion with velocity 10 m/sec after 8 sec its velocity becomes 24 m/sec then find total displacement in this interval. (a = constant)

131. Object starts his motion with velocity 10 m/sec and uniform acceleration 2m/s^2 then find velocity and displacement in 5 sec.

132. Object starts his motion from rest and acceleration = 5 m/s^2 then find displacement in 4 sec and in 4th sec.

133. Object starts his motion from rest and constant acceleration then find ratio of displacement in 6th-sec and 6-sec

134. Object starts his motion from rest and constr acceleration 'a' then find ration of displacement in 1sec, 2sec, 2sec, 4 sec,

135. If object starts his motion from rest and constant acceleration a then its displacement and distance will be same?

(*a*) True (*b*) False

136. If $u = 0$ (initial velocity) and constant acceleration then find ratio of distance in 4-sec and 8-sec.

137. Object starts his motion from rest and constant acceleration then find ration displacement in of 5sec and in next 5sec

138. Object starts from rest and constant acceleration it moves 80 m in 6-sec then find displacement in 12-sec

139. Object starts his motion from rest and constant acceleration then find ratio of displacement in 1st s: 2nd s: 3rd s

140. If velocity of object is given as $V = \sqrt{25 - 8x}$, then find displacement of particle in 1st second of it's motion.

141. A particle experiences a constant acceleration for 20 sec after starting from rest. If it travels a distance S_1 in the first 10 sec and a distance S_2 in the next 10 sec, then:

(*a*) $S_1 = S_2$ (*b*) $S_1 = S_2/3$

(*c*) $S_1 = S_2/2$ (*d*) $S_1 = S_2/4$

142. A body starts from rest. What is the ratio of the distance travelled by the body during the 4th and 3rd second.

143. Object starts his motion from rest and constant acceleration and moves 100 m in 1st 5-sec then find displacement in next 5-sec.

144. An object accelerates from rest to a velocity 27.5 m/s in 10 sec then find distance covered by object in next 10 sec.

(*a*) 550 m (*b*) 137.5 m

(*c*) 412.5 m (*d*) 275 m

145. Object starts his motion from u and constant acceleration then find velocity at mid point if velocity at end point is v.

146. Object starts his motion with u and constant acceleration a then find its velocity V_0 at one 3rd displacement of complete journey if final velocity is V.

147. A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 km/h and 40 km/h respectively. The velocity of the car midway between P and Q is

(*a*) 33.3 km /h (*b*) $20\sqrt{2}$ km / h

(*c*) $25\sqrt{2}$ km / h (*d*) 35 km /h

148. A car moving with a speed of 50 km/h can be stopped by applying brakes after at least 6 m. If the same car is moving with a speed of 100 km/h, what is the minimum stopping distance?

(*a*) 6 m (*b*) 12 m (*c*) 18 m (*d*) 24 m

149. A car moving with a speed of 40 km/h can be stopped by applying brakes after at least 2 m. If the same car is moving with a speed of 20 km/h, what is the minimum stopping distance?

(*a*) 4 m (*b*) 6 m (*c*) 8 m (*d*) 0.5 m

150. A particle start moving from rest and constant acceleration. It travels a distance x in first 2s, and y in 4s then correction relation is

(*a*) $y = 3x$ (*b*) $y = 4x$ (*c*) $y = x$ (*d*) $x = 4y$

151. A particle starts from rest and constant acceleration it moves 40 m in 3 sec then find distance in next 3sec[or] 3 sec to 6 sec.

152. Object starts his velocity 10 m/s/and acceleration 3 m/s² then find displacement between 4 sec to 8 sec.

153. A body starts from rest travelled a distance 120 m in the 8th sec then acceleration is:

(*a*) 10 (*b*) 8 (*c*) 16 (*d*) 4

154. A body starts from rest with constant acceleration, it moves S_1, S_2 and S_3 distance in 1st, 2nd and 3rd sec of its journey then correct relation is:

(*a*) $S_1 = S_2 = S_3$ (*b*) $5S_1 = 3S_2 = S_3$

(*c*) $S_1 = \frac{1}{3} S_2 = \frac{1}{5} S_3$ (*d*) $S_1 = \frac{1}{5} S_2 = \frac{1}{3} S_3$

155. Object loose its half velocity after travelling a distance 3 m with constance retardation, then find further distance after which it will stop.

(*a*) 3 m (*b*) 1 m (*c*) 2 m (*d*) 4 m

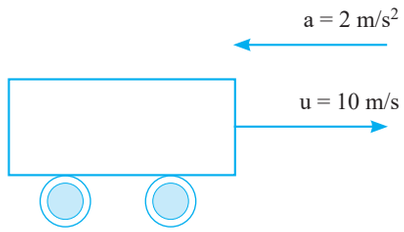
156. A body is moving with uniform acceleration describes (40 m in the first 5 sec and (65 m in next 5 sec) Its initial velocity will be

(*a*) 4 m/s (*b*) 2.5 m/s

(*c*) 5.5 m/s (*d*) 11 m/s

157. A particle moving with constant acceleration travels 24 m and 64 m in the first two consecutive interval of 4 sec its initial velocity is:

158. Find distance travelled in 7 sec if its motion is given



159. The initial velocity of particle is 20 m/s and retardation 2 m/s then find distance and displacement in 12. sec.

160. The initial velocity of particle is 20 m/s and retardation 2 m/s² then find distance and displacement in 10th sec.

161. The initial velocity is 21 m/s and retardation 2 m/s² then distance and displacement in 11th sec.

(*a*) 0,0 (*b*) 0.5.0

(*c*) 0,0.5 (*d*) 0.5 m 0.5 m

162. Object starts his motion from rest and constant acceleration takes time T for s displacement then find time taken for 1st half and 2nd half displacement

163. Two balls A and B are placed at the top of 180 m tall tower. Ball A is released from the top at $t = 0$ s. Ball B is thrown vertically down with an initial velocity v at $t = 2$s After a certain time, both balls met 100 m above the ground. Find value of V in ms^{-1}.

(use g = 10m/s²

(*a*) 10 (*b*) 15 (*c*) 20 (*d*) 30

164. An engine of a train, moving with uniform acceleration passes the signal post with velocity u and last compartment with velocity v. The velocity with which middle point of train passes signal post is:

(*a*) $\frac{u+v}{2}$ (*b*) $\frac{v-u}{2}$

(*c*) $\sqrt{\frac{v^2-u^2}{2}}$ (*d*) $\sqrt{\frac{v^2+u^2}{2}}$

Rest to Rest Motion

165. Object starts his motion from rest and constant acceleration α for some time t_1; then after retards with β and comes to at rest in time t_2 then find total distance travelled and V_{max}

166. A body starts from rest with acceleration 2 m/s² till it attains the maximum velocity then retards to rest with 3 m/s². If total time is 10 sec then maximum speed.

167. The engine of a motor cycle can produce maximum acceleration 5 m/s² It can produce maximum retardation 10 m/s² what is then minimum time in which he will cover a distance of 1.5 km

(*a*) 5 sec (*b*) 15 sec

(*c*) 10 sec (*d*) 30 sec

168. A particle starts from rest, accelerates at 2 m/s² for 10s and then goes with constant speed for 30s and then decelerates at 4 m/s² till it stops. What is the distance travelled by it

(*a*) 750 m (*b*) 800 m

(*c*) 700 m (*d*) 850 m

169. A car starts from rest and moves with uniform acceleration a on a straight road from time $t = 0$ to $t = T$. After that, a constant deceleration brings it to rest. In this process the average speed of the car is

(*a*) aT (*b*) $\frac{aT}{4}$ (*c*) $\frac{3aT}{2}$ (*d*) $\frac{aT}{2}$

170. A car accelerates from rest at a constant rate a for some time after which it decelerates at constant rate β and comes to rest. If total time elapsed is t. After how much time particle has attained it's maximum velocity?

(*a*) $\frac{t}{2}$ (*b*) $\frac{\alpha t}{\alpha+\beta}$ (*c*) $\frac{\beta t}{\alpha+\beta}$ (*d*) t

171. A particle of mass m moves on the x-axis as follows: It starts from rest at t = 0 from the point $x = 0$ and comes to rest at $t = 1$ at the point $x = 1$ No other information is available about its motion at intermediate time $[0 < t < 1]$ If a denotes the instantaneous acceleration of particle, then

(*a*) $|\alpha|$ cannot exceed 2 at any point in its path.

(*b*) $|\alpha|$must be > 4 at some point or points in its path

(*c*) $|\alpha|$cannot remain positive for all t in the interval $0 \le t \le 1$

(*d*) $|\alpha| = 2$ at any point in its path.

Motion under Gravity

172. Object is dropped & moved 75 m in last sec of it's Journey then find total time of journey

173. Object is dropped and moves 135 m in last 1 sec of its Journey then find time of flight.

174. Object is dropped and moves 185 m in last 1 sec of its Journey then find time of Flight & height from it is droid.

175. Object is dropped and moves 55 m in last 1 sec of its journey then find time of flight.

176. Motion under gravity is an example of

(*a*) on-uniform acceleration, uniform motion

(*b*) Non-uniform motion, Non-uniform acceleration

(*c*) Non-uniform motion, uniform acceleration

(*d*) Uniform motion, uniform acceleration

177. Ball is dropped and move 85 m in $\underline{n}^{th}$sec then find that time interval.

178. Ball is Projected with 50 m/s then find displacement in 3-sec and 3rd sec.

179. Ball is dropped from 125 m then distance moved in last 2 sec of Journey.

180. Ball is dropped then find ratio of distance in 3rd sec and 7th sec

181. water drops are falling from a nozzle of shower onto the floor, from a height of 9.8 m. The drops fall at regular interval of time. When the first drops strikes the floor, at that instant, the third drop begins to fall. Locate the position of second drop from floor when first drop strikes the floor

(*a*) 4.18 m (*b*) 2.94 m

(*c*) 2.45 m (*d*) 7.35 m

182. If velocity of object is given as $V = \sqrt{25 - 8x}$, then find displacement of particle in 1st second of it's motion.

183. Object is dropped from Height 180 m from ground find its velocity at ground. and total time of flight:

184. Ball is dropped and touches the ground in 4 -sec. then find height of towers from it dropped, velocity at ground, and distance traveled in last-sec of its journey.

185. A ball is dropped then find distance in 5th sec of journey

186. A Ball is dropped from a tower of Height 20m. then find velocity at ground.

187. Object is dropped and distance in last 1 sec is equal to 1st 3 sec then find height from ground from where ball is dropped.

188. A particle is dropped under gravity from rest from a height h and it travels a distance $\left[\frac{9h}{25}\right]$ in the last second the height h is

189. Ball is dropped from Height H and it travels $\frac{7H}{16}$ in last sec of Journey then find 'H'

190. A Ball is dropped at $t = 0$ sec after 1 sec, 2nd Ball is dropped after 2 sec, 3rd Ball is dropped, after 3 sec 4th Ball is dropped. then Find distance b/w 2nd and 3nd Ball when 4nd Ball is about to fall.

191. Water drop is falling in a regular intervals when 1st drop is reaches to ground then 4th drop is about to release, then find distance between 2nd drop and 3rd drop height of water top is 9 m

192. Water drops falls at regular interval from a tap which is 5m above the ground. The 3rd leaving the tap at the instant the 1st drop touches the ground. How far above the ground is the second drop at that instant.

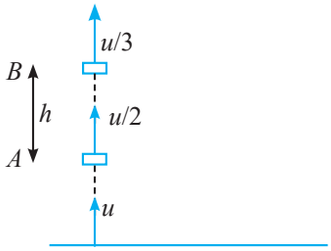
193. Ball is dropped from Height H then find ratio of time in 1st half and 2nd half distance.

194. Object is dropped then it moves 2nd half distance in last 1 sec of motion then find time of flight.

195. Object is projected with 40 m/s and 60 m/s respectively then find ratio of distance in last-sec of upward Journey

196. Ball is dropped from height H and takes T time to reach ground find Position of ball from ground after T/3 time.

197. Ball is Projected with speed u as shown in fig. then find distance between A & B



198. A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is: (Given = g = 9.8 m/s²)

(*a*) 5 m (*b*) 10 m

(*c*) 19.8 m (*d*) 24.5 m

199. Object is dropped then it moves 2nd half distance in last 1 sec of motion then find time of flight.

200. If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is

(*a*) ut (*b*) $\frac{1}{2}gt^2$

(*c*) $ut-\frac{1}{2}gt^2$ (*d*) $(u+gt)t$

201. Rocket starts his motion in upward direction with acceleration (10 m/s²) upward. After 5 sec engine off then find maximum height from ground.

202. From the top of a building, 16 m high water drops are falling at equal intervals of time such that when the first drop reaches the ground, the fifth drop just starts. Find the distance between the successive drops at that instant

203. An elevator in which a man is standing is moving upwards with a constant acceleration of 1m / s^2 At some instant when speed is (10 m/s). If the man drops a coin a from a height 2m, it reaches the floor of the elevator after a time (g = 10m/s^2)

(*a*) 2 sec (*b*) $\frac{2}{11}$sec

(*c*) $\sqrt{\frac{4}{11}}$sec (*d*) 1sec

204. Ball is dropped from height 125 m after 3 sec it stopped and released at same instant find Total time of flight.

205. Ball is Projected up with 80 m/s then find distance travelled in last sec of Journey.

(*a*) 35 m (*b*) 45 m (*c*) 80 m (*d*) 75 m

206. Ball is projected with speed 40 m/s then find

(*i*) H_{max}

(*ii*) T_f

(*iii*) T_{upword}

(*iv*) Speed after t = 5sec, 6 sec

(*v*) Distance in 6 sec

(*vi*) Displacement in 6 sec.

(*vii*) Avg speed in 7 sec

(*viii*) Avg velocity in 5 sec

(*ix*) Distance moved in 8th sec

207. Ball is projected upward with velocity 50 m/s then find :

(*i*) Time of flight

(*ii*) Maximum height

(*iii*) displacement in 6 sec

(*iv*) Height at 7 sec

(*v*) Average velocity in 8 sec

(*vi*) Average speed in 8 sec

208. Object is Projected with 80 m/s then find average speed velocity in 8-sec.

209. A Ball is thrown upward with speed 40 m/s then Find Avg. Velocity of upward Journey and speed at half of the maximum Height.

210. A Ball is thrown upward with u_0 if its velocity at half of maximum Height is 20 m/s then find it velocity u_0

211. A ball is thrown vertically upward with an initial velocity of 150 m/s. The ratio of velocity after 3s and 5s is $\frac{x+1}{x}$. The value of x is x (take g = 10 m/s^2)

(*a*) –5 (*b*) 10 (*c*) 5 (*d*) 6

212. Ball is Projected up with 50 m/s then Find distance moved in 8 -sec

213. Ball is Projected up with 70m/s then find displacement in 10th sec and 10 sec.

214. Ball is Project up with 45 m/s then find distance moved in 5th sec .

215. Ball is projected up its position at t = 7s and t = 11s is same then, find velocity of projection and maxm height

216. Object is Projected up with u, its height at 3 sec and 13 sec is same, then find u and that Height,

217. Ball is projected up its distance traveled in 3rd sec and 8th sec is same then find maximum height

218. Bal is projected up with 55m/s then find distance in 6th sec.

219. Object is projected with 40 m/s then find average speed and velocity in 6 sec.

220. Object is projected in upward with velocity 75 m/s. Find distance in last 2 sec of upward journey.

221. Ball is projected with speed u it crosses the height h in time t_1 and t_2 then find that height.

222. Ball is projected with speed 50 m/s. Then, find displacement in last 3 sec of journey.

(*a*) 45 m (*b*) 35 m
(*c*) 105 m (*d*) 80 m

223. Ball is projected with speed u_1 if air friction is not ignored then speed of collision u_2 and t_1 and t_2 are time of upward and downward time, then

(*a*) $t_1 = t_2, u_1 > u_2$ (*b*) $t_1 < t_2, u_1 > u_2$
(*c*) $t_1 < t_2, u_1 < u_2$ (*d*) $t_1 > t_2, u_1 > u_2$

224. If constant air friction is acting on object is a_0 then find ratio of time of flight of upward and downward Journey.

Motion under Gravity Height to Ground

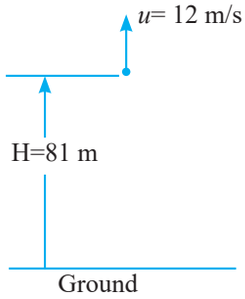
225. A ball is dropped from a high rise platform at $t = 0$ starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at t = 18 s. What is the value of v? (Take g = 10m / (s^2)

(*a*) 75 m /s (*b*) 55 m /s
(*c*) 40 m /s (*d*) 60 m /s

226. Ball is Projected up with speed 4 m/s from a bridge The Ball strikes the water surface after 4 s The Height of Bridge above the water surface

(*a*) 80 m (*b*) 100 m (*c*) 60 m (*d*) 40 m

227. Find time of *t* light of ball



(*a*) 2.7 sec (*b*) 5.4 sec
(*c*) 4.8 sec (*d*) 6.9 sec

228. Ball is projected up with speed "u" from height H. Then time of fight T_1. With same speed "u" it is projected downward then time of fight is T_2. find time of fight "T" when object is dropped from same height.

229. A ball is projected up with speed 20 m/s from a building height 25 m. Then, find total time to reach the ball on ground.

230. Ball is projected from height of 75 m in upward direction with speed 10 m/s then find

(*i*) Total time of flight.
(*ii*) Velocity at ground.
(*iii*) Maximum height from ground.

231. Ball is projected up from height *H* with speed *u* and collide with 3*u* at ground then find *H* and time of flight

232. A ball is projected up with speed u from height H and it takes 9s to reach ground another same ball projected downward with same speed it takes 4s to reach ground if another ball is dropped from same height. Then, it will take to reach ground.

233. A person standing on the floor of an elevator drops a coin. Coin reaches the floor in time t_1 if the elevator is moving uniformly and time t_2 if elevator is stationary. Then :

(*a*) $t_1 < t_2$ or $t_1 > t_2$ depending upon whether lift is going up or down.
(*b*) $t_1 < t_2$
(*c*) $t_1 > t_2$
(*d*) $t_1 = t_2$

234. A lift is moving with uniform speed 20 m/s. Ball is released from lift when lift is at height 60 m from ground. Find total time of flight

235. A balloon starts raising up from ground with 1.25 m/s² after 8 sec a small particle is dropped then find time of flight of particle

236. A balloon was moving upwards with uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from ground level. The height of balloon from ground when object strikes the ground was around

(*a*) 300 m (*b*) 125 m
(*c*) 200 m (*d*) 250 m

Juggler Problem

237. A Juggler wants to Keep *n* Ball in air, if he is throwing Ball with speed *u* then what shout be time interval

238. A Juggler maintain 4 Ball in air with throwing speed 20 m/s then find time interval, and Position of Balls when one ball Just about to Project.

239. A juggler maintains four balls in the air with air with throwing speed 20 m/s upwards in regular time intervals. When one ball is about to leave his hand the height of balls in air from the ground will be:

(*a*) 60 m, 80 m, 60 m, 0 m (*b*) 30 m, 40 m, 30 m, 0 m
(*c*) 15 m, 20 m, 15 m, 0 m (*d*) 10 m, 20 m, 10 m, 0 m

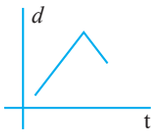
240. Juggler wants to keep 5 Ball in air he project each ball with a time interval of 2 sec. Then find velocity of projectile.

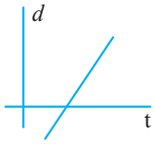
241. A juggler maintains 10 balls in motion, making each of them rise at a height of 80 m from his hands. Find the time interval maintained by the juggler to keep the proper distance between them

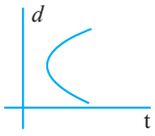
(*a*) 0.6 s (*b*) 0.8 s
(*c*) 1.0 s (*d*) 1.2 s

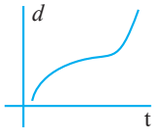
Graph based Questions

242. Which of the following graph is correct for distance-time

(a) 

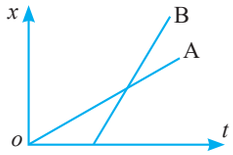
(b) 

(c) 

(d) 

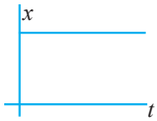
243. The position-time graphs for two students A and B returning from the school to their homes are shown in figure.

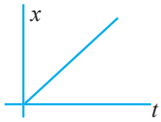
(*i*) A lives closer to the school
(*ii*) B lives closer to the school
(*iii*) A takes lesser time to reach home
(*iv*) A travels faster than B
(*v*) B travels faster than A.

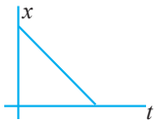


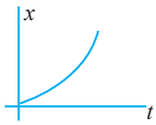
(a) (*i*), (*iii*), (*iv*) only (b) (*i*), (*v*) only
(c) (*i*), (*iii*) and (*v*) only (d) (*ii*) and (*v*) only

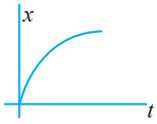
244. Comment nature of motion for given graph?

(*i*) 

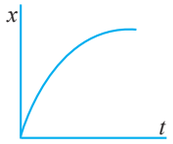
(*ii*) 

(*iii*) 

(*iv*) 

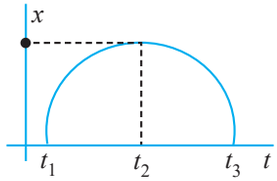
(*v*) 

245. Acceleration for given position-time graph is



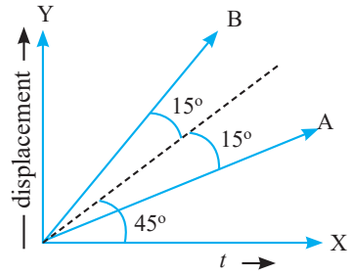
(a) -ve (b) +ve
(c) zero (d) increasing

246. In which time interval acceleration & velocity is parallel



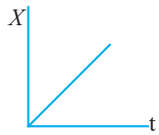
(a) t_1 to t_2 (b) t_1 to t_3 (c) t_2 to t_3 (d) None of this

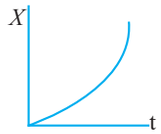
247. The displacement-time graph for two particles A and B is follows. The ratio V_A/V_B is

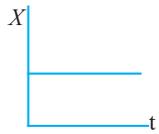


(a) 1 : 2 (b) 13 : 1
(c) $1:\sqrt{3}$ (d) 1 : 3

248. In which graph acceleration is non - zero constant?

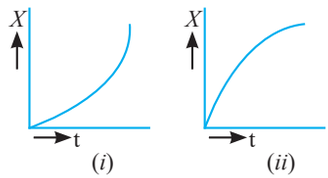
(a) 

(b) 

(c) 

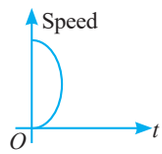
(d) None of these

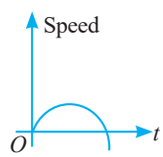
249. Figures (*i*) and (*ii*) below show the displacement-time graphs of two particles moving along the x-axis. We can say that

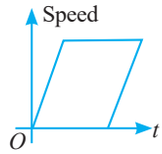


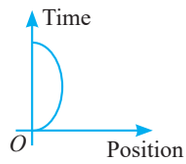
(a) Both the particles are having a uniformly accelerated motion
(b) Both the particles are having a uniformly retarded motion
(c) Particle (*i*) is having a uniformly accelerated motion while particle (*ii*) is having a uniformly retarded motion
(d) Particle (*i*) is having a uniformly retarded motion while particle (*ii*) is having a uniformly accelerated motion

250. Which one of the following graph for a body moving along a straight line is possible?

(a) 

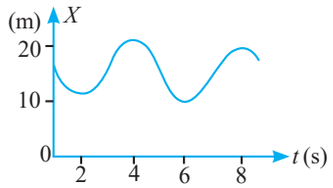
(b) 

(c) 

(d) 

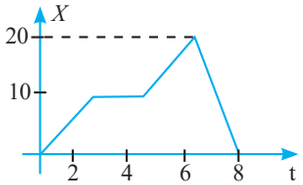
251. **Statement-I:** Area under velocity-time graph gives distance travelled by body in given time.
Statement-II: Area under acceleration-time graph is equal to change in velocity in given time.
(*a*) Both statements false
(*b*) Both statements true
(*c*) Statement-I is false and Statement-II is true/
(*d*) Statement-l is true and Statement-II is false

252. Figure shows the position of a particle moving on the x-axis as a function of time

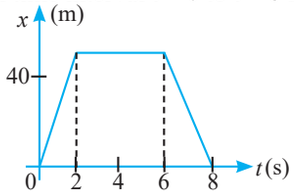


(*a*) The particle has come to rest 4 times
(*b*) The velocity at t = 8 s is negative
(*c*) The velocity remains positive for t = 2 s tot = 6s
(*d*) particle moves with a constant velocity

253. Find average speed and velocity in 8 sec.

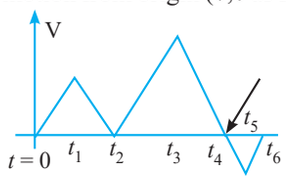


254. The position (*x*) of a particle moving along x-axis varies with time (t) as shown in figure. The average acceleration of particle in time interval t = 0 to t = 8 s is



(*a*) 3 m/s^2 (*b*) –5 m/s^2 (*c*) –4 m/s^2 (*d*) 2.5 m/s^2

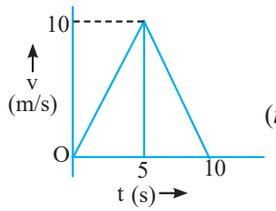
255. Object starts motion from origin (0,0 as shown in the figure)



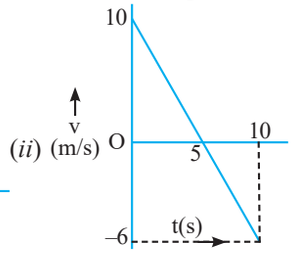
(*i*) Then its position/ displacement is maximum at:
(*ii*) Then its velocity is maximum at:

256. A particle is moving in a straight line. Its *v*-*t* graph in different cases are as shown below. Find the displacement and distance in time interval *t* = 0 to *t* = 10s for the following cases:

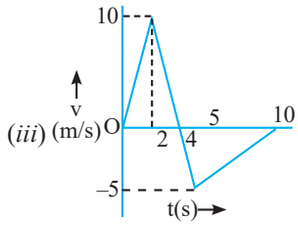
(*i*)



(*ii*)

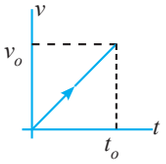


(*iii*)

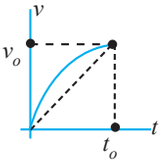


257. In which case average Velocity is maximum?

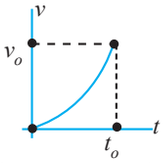
(*a*)



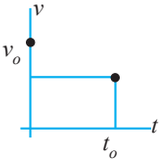
(*b*)



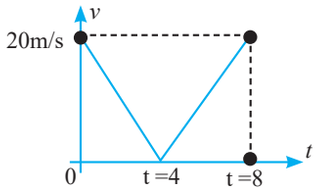
(*c*)



(*d*)

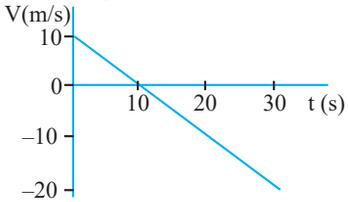


258. Find average acceleration in 8 sec.



(*a*) Zero (*b*) +5m/s^2
(*c*) -5m/s^2 (*d*) all of them

259. The velocity-time plot for a particle moving on a straight line shown in the figure



(*a*) The particle has a constant acceleration
(*b*) The particle has never turned around
(*c*) The particle has a zero displacement
(*d*) The average speed in the interval 0 to 10s is the same as the average speed in the interval 10s to 30s

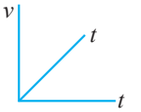
260. Match the following:

(*a*)



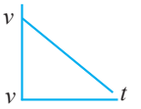
(*i*) Retardation

(*b*)



(*ii*) Uniform motion

(*c*)



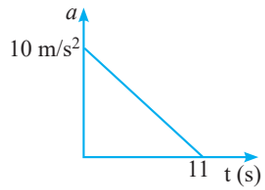
(*iii*) Constant acceleration

(*d*)



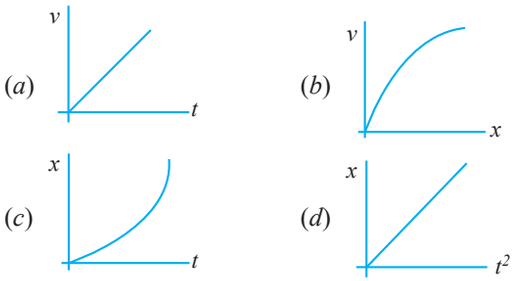
(*iv*) Rest

261. A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be



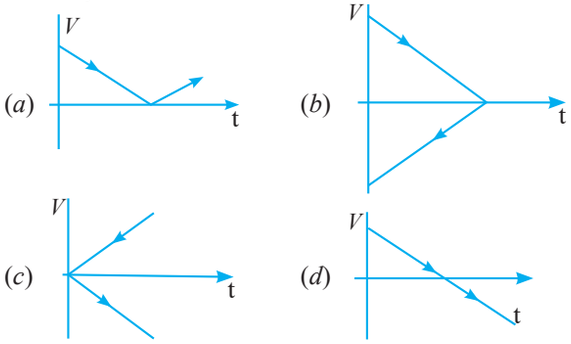
(a) 110 m/s (b) 55 m/s (c) 550 m/s (d) 660 m/s

262. In which case acceleration is constant.

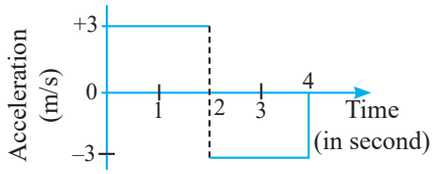


(e) In all

263. A body is thrown vertically upwards. Which of the following graphs represents the velocity of the body against time correctly?

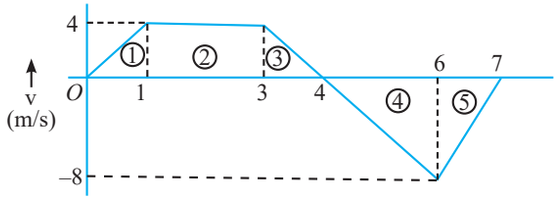


264. A particle starts from rest at t=0 and moves in a straight line with an acceleration as shown below. The velocity of the particle at t = 3s is

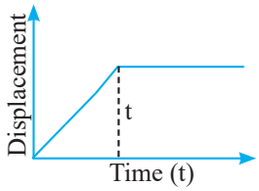


(a) 2 m/s (b) 3 m/s (c) 4 m/s (d) 6 m/s

265. A particle is moving in a straight line along the x-axis. Its velocity-time graph is as shown. Sketch the displacement-time and distance-time graphs.

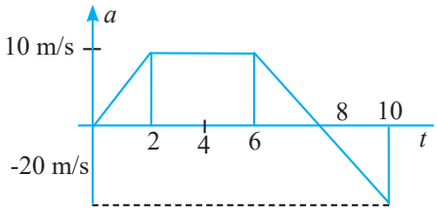


266. The x-t graph shown in figure represents

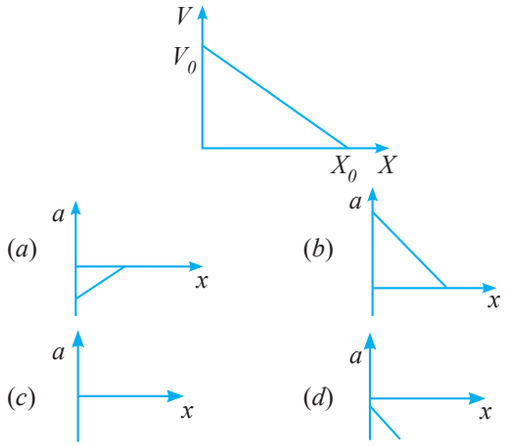


(a) Constant velocity.
(b) Velocity of the body is continuously changing
(c) Instantaneous velocity is zero at t = 0.
(d) The body travels with constant speed upto time t, and then stops

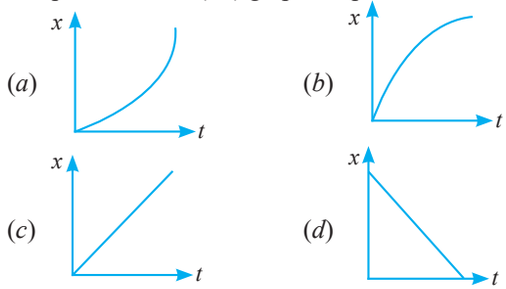
267. If initial velocity is –10 m/s then find velocity at t = 10 sec



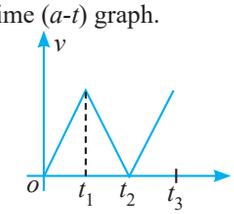
268. The given figure graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement



269. The position-time (x-t) graph for positive acceleration is:

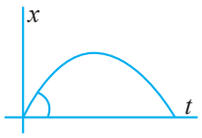


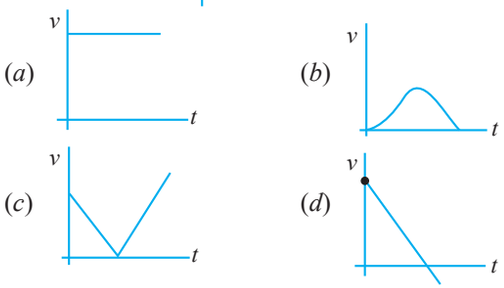
270. The velocity-time (v-t) graph of a particle is shown below. Convert the given velocity-time (v-t) graph into an acceleration-time (a-t) graph.



271. The motion of a body is represented by a position-time (x-t) graph, where the body starts with an initial positive velocity, reaches a maximum displacement, and then moves back towards its starting point with decreasing velocity. The graph exhibits different slopes, indicating changes in velocity.

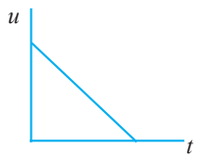
Convert the given position-time (x-t) graph into a velocity-time (v-t) graph.

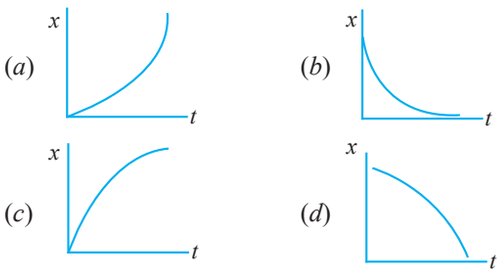




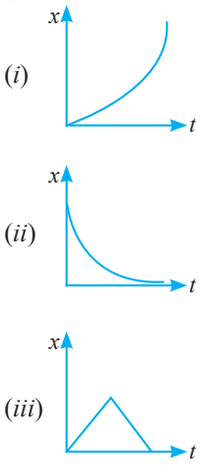
272. The velocity-time (v-t) graph of a body moving under uniform retardation is shown below.

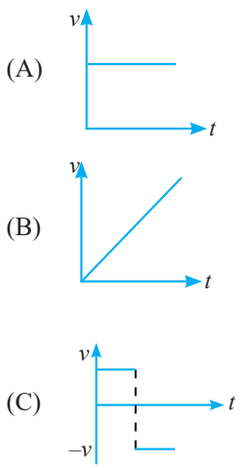
Convert the given velocity-time (v-t) graph into a position-time (x-t) graph.

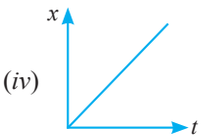


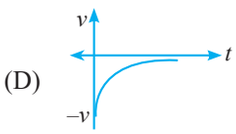


273. Match the column I with II and choose the correct option given below.



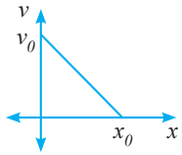


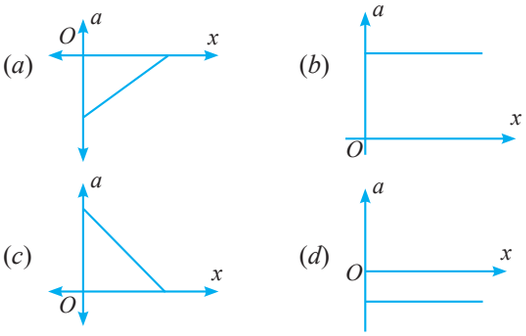




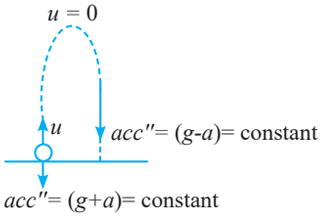
(*a*) (*i*)-B, (*ii*)-D, (*iii*)-C, (*iv*)-A
(*b*) (*i*)-A, (*ii*)-B, (*iii*)-C, (*iv*)-D
(*c*) (*i*)-A, (*ii*)-D, (*iii*)-B, (*iv*)-C
(*d*) (*i*)-D, (*ii*)-B, (*iii*)-C, (*iv*)-A

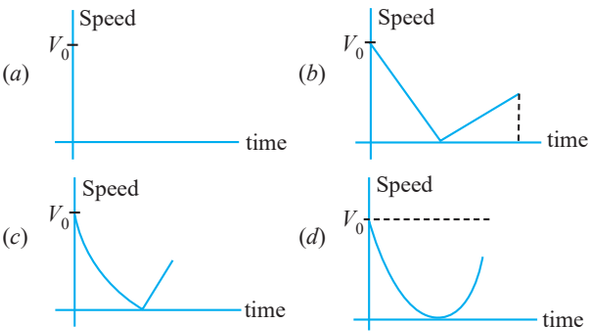
274. The velocity-displacement graph of particle is shown in figure. The acceleration-displacement graph of same particle is represented by



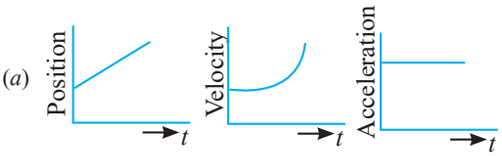


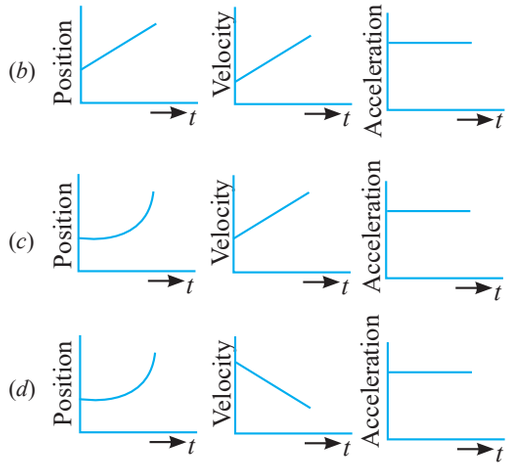
275. A ball is thrown vertically upwards. Which of the following plots represents the speed-time graph of the ball during its flight if the air resistance is not ignored? acceleration due to air friction = a_0



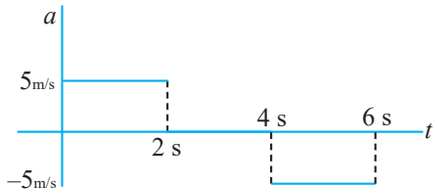


276. The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by:





277. A particle starts from rest and moves under uniform acceleration. The velocity-time graph for the motion is shown below.



Draw the velocity-time graph for the particle's motion if the initial velocity is zero.

Relative Velocity

278. Two objects, A and B, are moving in the same direction along a straight line. The velocity of object A is 4 m/s, and the velocity of object B is 10 m/s. Find the relative velocity of B with respect to A.

279. Two objects, A and B, are moving in the same direction along a straight line. The velocity of object A is 15 m/s, and the velocity of object B is 54 km/h. Find the relative velocity of A with respect to B

280. A train of 150m length is going towards north a speed of 10 m/s. a birds is flying to the track towards south. the time bird to cross the train is

(*a*) 10 s (*b*) 15 s (*c*) 30 s (*d*) 12 s

281. Two trains each of length 100 m moving parallel towards each other at speed 72 km/h and 36 km/h respectively. In how much time will they cross each other?

(*a*) 4.5 s (*b*) 6.67 s (*c*) 3.5 s (*d*) 7.25 s

282. A bus is moving with a speed of 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus?

(*a*) 40 ms^{-1} (*b*) 25 ms^{-1} (*c*) 10 ms^{-1} (*d*) 20 ms^{-1}

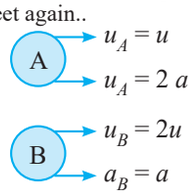
283. Two cars A and B are moving in same direction with velocities 30 m/s and 20 m/s. When car A is at a distance d behind the car B, the driver of the car A applies brakes producing uniform retardation of 2 m/s^2. There will be no collision when

(*a*) $d < 2.5\ m$ (*b*) $d > 125\ m$
(*c*) $d > 25\ m$ (*d*) $d < 125\ m$

284. Two trains each of length 50 m are approaching each other on parallel rails. Their velocities are 10 m/sec and 15 m/sec. They will cross each other in-

(*a*) 2 sec (*b*) 4 sec (*c*) 10 sec (*d*) 6 sec

285. If both objects start from the same position at $t = 0$, find the time when they will meet again..



286. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be

(*a*) $\frac{t_1 + t_2}{2}$ (*b*) $\frac{t_1 t_2}{t_2 - t_1}$ (*c*) $\frac{t_1 t_2}{t_2 + t_1}$ (*d*) $t_1 - t_2$

287. Two cars are moving in the same direction with a speed of 30 km/h. They are separated from each other by 5 km. Third car moving in the opposite direction meets the two cars after an interval of 4 minutes. The speed of the third car is

(*a*) 30 km/h (*b*) 25 km/h
(*c*) 40 km/h (*d*) 45 km/h

288. A ball thrown downward with speed 20 m/s and 30 m/s simultaneously, then find relative velocity and separation between them after 4 sec

289. A passenger sitting in a train A moving at 90 km/h observes another train B moving in the opposite direction for 8 seconds. If the velocity of the train B is 54 km/h, then length of train B is:

(*a*) 80 m (*b*) 200 m (*c*) 120 m (*d*) 320 m

290. Two cars A and B are travelling in the same direction with velocities V_1 and V_2 [$V_1 > V_2$]. When the car A is at a distance *d* ahead of car B, the driver of the car A applied the brake producing a uniform retardation a. There will be no collision when

(*a*) $d < \frac{(V_1 - V_2)^2}{2a}$ (*b*) $d < \frac{V_1^2 - V_2^2}{2a}$

(*c*) $d > \frac{(V_1 - V_2)^2}{2a}$ (*d*) $d > \frac{V_1^2 - V_2^2}{2a}$

291. Two trains 'A' and 'B' of length '*l*' and '4*l*' are travelling into a tunnel of length 'L' in parallel tracks from opposite directions with velocities 30 m/s and 20 m/s respectively. If train 'A' takes 35 s less time than train 'B' to cross the tunnel, then length 'L' of found is: (given L = 60*l*)

(*a*) 900 m (*b*) 1200 m (*c*) 1800m (*d*) 2700 m

292. A police jeep is chasing with, velocity of 45km/ka thief in another jeep moving with velocity 153km/h. Police fires a bullet with muzzle velocity of 180m/s. The velocity it will strike the car of the thief is.

(*a*) 150 m/s (*b*) 27 m/s
(*c*) 450 m/s (*d*) 250 m/s

Answer Key

1. (*i*) 5 m, (*ii*) 5m, (*iii*) 25 m, (*iv*) $4\sqrt{2}m$ **2.** (*e*) **3.**(*a*) Must not (*b*) May (*c*) May/may not (*d*) Must not **4.** (*b*)
5. (*b*) **6.** (*a*) **7.** (*a*) **8.** (*a*) **9.** (*b*) **10.** (*c*) **11.** (*a*) **12.** (*b*) **13.** (*d*)
14. (*i*)- D, (*ii*)- C, (*iii*)- B, (*iv*)- B **15.** (*d*) **16.** (*d*)
17. Distance → impossible to calculate form given data, displacement = $10\sqrt{2}$ **18.** Not able to find distance, $5\sqrt{2}m$
19. (*a*) 32 km, (*b*) 8 km **20.** (*i*) $\frac{\pi}{3}$, (*ii*) $\frac{3\pi}{2\sqrt{2}}$, (*iii*) $\frac{2\pi}{3\sqrt{3}}$ **21.**1-a, 2 - d, 3 - b, 4 - c and 1-iii, 2-iv, 3- i, 4 - ii
22. 1-d, 2-c, 3 - a, 4 -c, 5 - d, 6 - b **23.** 1-f, 2-b, 3 - e, 4 -d, 5 - a, 6 - c **24.** 0 **25.** Position = 11m, displacement = 8 m
26. 10 m, west **27.** $10\sqrt{3}$ **28.** (*c*) **29.** $\sqrt{3}l$ **30.** $\sqrt{5}l$ **31.** (*i*)$\sqrt{\frac{40}{9}}$ (*ii*) $2\times\sqrt{\frac{40}{9}}$ (*iii*) $2\sqrt{2}$ft **32.** (*a*)
33. (*i*) 20 m (*ii*) 0 m **34.** (*i*) 20 m (*ii*) 20 m **35.** Distance = 35 m, Displacement = 35 m **36.** (*b*) **37.** $\sqrt{13}$m **38.** (*a*)
39. $\frac{x_1+x_2}{t_1+t_2}$ **40.** Average velocity = $\frac{x_1-x_1}{t_1+t_2}$, Avg. Speed = can't be calculated **41.** (*d*) **42.** (*a*) **43.** (*d*) **44.** (*b*)
45. 25 m/s **46.** 29.52 m/s **47.** 24 m/s **48.** 16.6 m/s **49.** 40 m/s, 0 **50.** (*b*) **51.** $\frac{v_1+2v_2}{3}$ **52.** $\frac{200}{3}m/s$
53. 20 m/s **54.** (*a*) **55.** Average speed: V, V, V, V, Average velocity: V, $\frac{V}{\sqrt{2}}$, V/3, 0 **56.** Average speed = instantaneous
57. (*d*) **58.** 2 m/s
59. A =Velocity may an may not be constant, B =Velocity must be variable, C = Speed must be constant D = Speed may or may not be constant **60.** t = 2.5s **61.** 4 m/s **62.** 9 m/s **63.** $2v/\pi$ **64.** (*a*)
65. $V\left\{\frac{\sin\left(\frac{\theta}{2}\right)}{\frac{\theta}{2}}\right\}$ **66.** (*b*) **67.** (*a*) **68.** (*c*)
69. (*b*) **70.** (*b*) **71.** (*a*) **72.** $\sqrt{3}:1$ **73.** (*b*) (*d*) (*e*) **74.** (*d*) **75.** (*a*) **76.** (*c*) **77.** (*b*)
78. (*c*) **79.** 18m/s **80.** 5 m/s^2 **81.** 40 m/s **82.** (*c*) **83.** (*b*) **84.** (*d*) **85.** (*c*) **86.** (*a*) **87.** $\vec{a}=\frac{\sqrt{2}}{5}m/s$
88. (*i*) $4\sqrt{2}$m/s^2, (*ii*) = 0 **89.** (*i*) = 0, (*ii*) 6m/s^2, (*iii*) $a = 12t$, (*iv*) $a = 12t^2$, (*v*) $a = 6t$, (*vi*) $a = 3$, (*vii*) $a = 3t^2$, (*viii*) $a = (4x + 8)$
90. $a=\frac{24}{t^4}$ **91.** (*i*), (*ix*), (*xii*) **92.** $x_i = \gamma$ m, $v_i = \beta$ m/s, $a_i = 2\alpha$ m/s^2 **93.** 10 m/s^2 **94.** (2k^2) m/s^2 **95.** 6 m/s^2, 12 m/s^2
96. (*i*) a = 2m/s^2, (*ii*)V = 0 m/s^2, (*iii*) 62 m/s^2, **97.** $\frac{a}{3b}$ **98.** 2s, 4m **99.** 8m **100.** (*b*) **101.** (*d*) **102.** (*b*)
103. (*c*) **104.** (*c*) **105.** $a=\frac{k^2}{2}$ **106.** (*b*) **107.** (*c*) **108.** (*c*) **109.** (*d*) **110.** t = 2 sec
111. $u_{(2)}$ = 8 m/s, $a_{(2)}$ = 16 m/s^2, $x_{(2)}$ = 4 m **112.**-27m/s^2 **113.** a = 24 m/s^2 **114.** (*a*)
115. (*d*) **116.** (*b*) **117.** (*a*) **118.** $\sqrt{\frac{8}{5}x^{5/2}+16}$ **119.** (*d*) **120.** Distance = displacement = 40 m **121.** 10 m, –8m
122. 2 m/s,0 **123.** (*b*) **124.** (*d*) **125.** (*c*) **126.** 240 m **127.** (*a*) **128.** (*d*) **129.** (*b*) **130.** 136 m
131. V = 20 m/s, s = 75 m **132.** 40 m, 17.5 m **133.**11 : 36 **134.**1 : 4 : 9 : 16 **135.** (*a*) **136.** 1 : 4
137. 1 : 3 **138.** 320 m **139.** 1 : 3 : 5 **140.** 3 m **141.** (*b*) **142.** 7 : 5 **143.** 300 m **144.** (*c*) **145.** $\sqrt{\frac{v^2+u^2}{2}}$
146. $V_o=\sqrt{\frac{v^2+2u^2}{3}}$ **147.** (*c*) **148.** (*d*) **149.** (*d*) **150.** (*b*) **151.** 120 m **152.** 112m **153.** (*c*) **154.** (*c*)
155. (*b*) **156.** (*c*) **157.** u = 1 m/s **158.** Distance = 29 meters **159.**Distance = 104 m, displacement = 96 m
160. 1 m **161.** (*c*)

162. $T_1 = \frac{T}{\sqrt{2}}, T_2 = T - \frac{T}{\sqrt{2}}$ **163.** (*d*) **164.** (*d*) **165.** $V_{max} = \left(\frac{\alpha.\beta}{\alpha+\beta}\right)(t_1+t_2), S = \frac{1}{2}\left(\frac{\alpha.\beta}{\alpha+\beta}\right)(t_1+t_2)^2$ **166.** 12 m/s

167. (*d*) **168.** (*a*) **169.** (*d*) **170.** (*c*) **171.** (*c*) **172.** 8 sec **173.** 14 sec **174.** T = 19 sec, H = 1805 m

175. 6 sec **176.** (*c*) **177.** $n = 9^{th}$ sec **178.** 105m, 25m **179.** 80m **180.** 15 : 39 **181.** (*d*)

182. 3m **183.** *v* = 60 m/s, 6 sec **184.** H = 80 m V = 40 m/s = $S_{last\ sec}$ = 35 m **185.** 45 m **186.** 20 m/s **187.** H = 125 m

188. 125 m **189.** 80 m **190.** 15 m **191.** 3 m **192.** 3.75 **193.** $t_1 : t_2 = 1 : \sqrt{2} - 1$ **194.** $T = \left(1 - \frac{1}{\sqrt{2}}\right)$sec **195.** 1 : 1

196. $8H/9$ **197.** $= \left[\frac{u^2}{144}\right]$ **198.** (*a*) **199.** $T\frac{\sqrt{2}}{\sqrt{2}-1} = \left(2+\sqrt{2}\right)$ **200.** (*b*) **201.** 250 m

202. *x* = 1m **203.** (*c*) **204.** 8 second **205.** (*d*)

206. (*i*) 80 m, (*ii*) 8 sec, (*iii*) 4 sec, (*iv*) 10m, 20m (*v*) 100 m, (*vi*) 60 m, (*vii*) $\frac{125}{7}$, (*viii*) $\frac{75}{5}$, (*ix*) 35 m

207. (*i*) 10 sec (*ii*) 125 m (*iii*) 120 m (*iv*) 105 m (*v*)10 m/s (*vi*) $\frac{170}{8} m/s$ **208.** 40 m/s **209.** 20m/s, $20\sqrt{2} m/s$

210. $20\sqrt{2} m/s$ **211.** (*c*) **212.** 170 m **213.** 200 m, –25 m **214.** 2.5 meters **215.** 90 m/s, 405 m

216. 80 m/s, 195 m **217.** 125 m **218.** 2.5 m **219.** 100/6 m/s, 10 m/s **220.** 20 m **221.** $\frac{1}{2} g t_1 \cdot t_2$ **222.** (*c*)

223. (*b*) **224.** $\frac{T_{up}}{T_{down}} = \sqrt{\frac{g-a_o}{g+a_o}}$ **225.** (*a*) **226.** (*d*) **227.** (*b*) **228.** $t = \sqrt{T_1 . T_2}$ **229.** 5 sec

230. 5 sec, 40m/s, 80 m **231.** $H = \frac{4u^2}{g}, T = \frac{4u}{g}$ **232.** 6 sec **233.** (*d*) **234.** 6 sec **235.** t_2 = 4 sec

236. (*b*) **237.** $t = \frac{2u}{ng}$ **238.** 1sec **239.** (*c*) **240.** 50 m/s **241.** (*b*) **242.** (*d*) **243.** (*b*)

244. (*i*) rest (*ii*) uniform motion, (*iii*) uniform motion in negative direction, (*iv*) Non-uniform (*v*) retardation

245. (*a*) **246.** (*c*) **247.** (*d*) **248.** (*b*) **249.** (*c*) **250.** (*d*) **251.** (*c*) **252.** (*a*) **253.** 5 m/s, 0 m/s

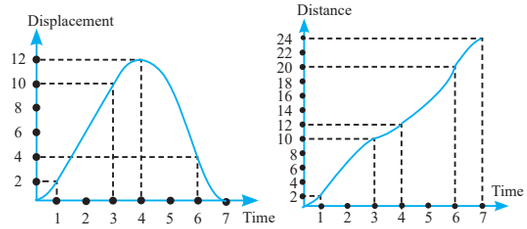
254. (*b*) **255.** (*i*) t_4, (*ii*) t_3

256. (*i*) Distance = displacement = 50 m, (*ii*) Distance = 50 m, displacement = 0, (*iii*) Distance = 35 m, displacement = 5m

257. (*a*) **258.** (*d*) **259.** (*a*) **260.** (A – iv, B – ii, C – iii, D – i) **261.** (*b*) **262.** (*e*) **263.** (*d*)

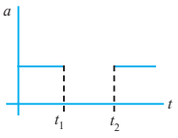
264. (*b*)

265.

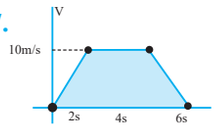


266. (*d*) **267.** u_f = 30 m/s **268.** (*a*) **269.** (*a*)

270.



271. (*d*) **272.** (*c*) **273.** (*a*) **274.** (*a*) **275.** (*b*) **276.** (*b*) **277.**



278. 6 m/s **279.** 0 **280.** (*a*) **281.** (*b*) **282.** (*d*) **283.** (*c*) **284.** (*b*) **285.** $T = \frac{2u}{a}$ **286.** (*c*)

287. (*d*) **288.** 40 m **289.** (*d*) **290.** (*c*) **291.** (*c*) **292.** (*a*)